Guided Notes: Module 6 – Implementing Network Services

Module Objectives

By the end of this module, you should be able to:

- Compare and contrast transport protocols.
- Use command line tools to scan network ports.
- Explain the use of network addressing services.
- Explain the use of name resolution services.
- Configure and troubleshoot DHCP and DNS services.

Lesson 6.1: Transport & Application Layer Protocols

Transport Layer Overview

- Responsible for multiplexing application data delivery.
- Provides basic connectivity, addressing, and forwarding.
- Common protocols: TCP (Transmission Control Protocol) and UDP (User Datagram Protocol).

Common TCP and UDP Ports and Their Uses

Port	Protocol	Service/Application	Description
20	TCP	ftp-data	File Transfer Protocol—Data
21	TCP	ftp	File Transfer Protocol—Control
22	TCP	ssh/sftp	Secure Shell / FTP over SSH
23	TCP	telnet	Telnet remote access
25	TCP	smtp	Simple Mail Transfer Protocol (email sending)
53	TCP/UDP	domain	Domain Name System (DNS)
67	UDP	bootps	BOOTP/DHCP Server
68	UDP	bootpc	BOOTP/DHCP Client
69	UDP	tftp	Trivial File Transfer Protocol
80	TCP	http	HyperText Transfer Protocol (web traffic)

110	TCP	pop	Post Office Protocol (email retrieval)
123	UDP	ntp/sntp	Network Time Protocol / Simple NTP
143	TCP	imap	Internet Message Access Protocol (email retrieval)
161	UDP	snmp	Simple Network Management Protocol
162	UDP	snmp-trap	SNMP Trap messages
389	TCP/UDP	ldap	Lightweight Directory Access Protocol
443	TCP	https	HTTP Secure (SSL/TLS encrypted web traffic)
445	TCP	smb	Server Message Block over TCP/IP (file sharing)
514	UDP	syslog	System logging
546	UDP	dhcpv6-client	DHCPv6 Client
547	TCP	dhcpv6-server	DHCPv6 Server
587	TCP	smtps	SMTP Secure (email sending over SSL/TLS)

636	TCP	ldaps	LDAP Secure
993	TCP	imaps	IMAP Secure
995	TCP	pop3s	POP3 Secure
1433	TCP	sql-server	Microsoft SQL Server
1521	TCP	sqlnet	Oracle SQL*Net
3306	TCP	mysql	MySQL / MariaDB
3389	TCP	rdp	Remote Desktop Protocol
5004	UDP	rtp	Real-Time Protocol (media streaming)
5005	UDP	rtcp	Real-Time Control Protocol
5060	TCP/UDP	sip	Session Initiation Protocol (VoIP signaling)
5061	TCP/UDP	sips	SIP Secure

TCP Handshake (Connection Establishment)

- TCP is connection-oriented, reliable.
- Three-way handshake process:
    1. Host A sends SYN (synchronize) to Host B.
    2. Host B responds with SYN + ACK (acknowledgment).
    3. Host A sends ACK back.
- After handshake, connection is established and communication can begin.

UDP Communication

- UDP is connectionless, no handshake.
- Data is sent as datagrams without acknowledgments.
- Faster but less reliable than TCP.
- Used where speed is prioritized over reliability (e.g., streaming, gaming).

Activity: Using netstat Command

- Open Command Prompt or Terminal.
- Run
- `netstat`
- to view active network connections and ports.
- Options:
    - `-a`
    - : Displays all active connections and listening ports.
    - `-p`
    - : Shows process IDs.
    - `-b`
    - : Displays executable involved (requires admin privileges).

Lesson 6.2: DHCP (Dynamic Host Configuration Protocol)

DHCP Process

- Enables automatic assignment of IP addressing and network configuration.
- Four-step process between client and server:
    1. **DHCP Discover**: Client broadcasts to find available DHCP servers.
    2. **DHCP Offer**: Server offers an IP address and configuration.
    3. **DHCP Request**: Client requests the offered IP address.
    4. **DHCP Acknowledgment (Ack)**: Server acknowledges and completes assignment.

DHCP Reservations and Exclusions

- Reservations: Assign a fixed IP to a specific client (based on MAC address).
- Exclusions: IP addresses excluded from the DHCP pool to avoid conflicts.

Activity: Identify the Protocol

- Review network traffic or packet captures and identify DHCP messages by sequence.

Lesson 6.3: APIPA and SLAAC

APIPA (Automatic Private IP Addressing)

- When DHCP fails, devices assign themselves an IP in the 169.254.x.x range.
- Allows limited local network communication without DHCP.

IPv6 Autoconfiguration

- Two main methods for IPv6 address assignment:
    1. **SLAAC (Stateless Address Autoconfiguration)**: Devices self-configure addresses using router advertisements, no server needed.
    2. **DHCPv6**: Similar to DHCP for IPv4, provides managed addresses.

Activity: Think About It

- For a dynamic, diverse campus network needing easy management, **SLAAC** is suitable for IPv6 autoconfiguration.

Lesson 6.4: DHCP Relay and Troubleshooting

DHCP Relay Agent

- Acts as a bridge between clients and remote DHCP servers.
- Forwards DHCP broadcast messages across different subnets.
- Allows centralized DHCP servers to serve multiple networks.

Network Diagram

- Shows DHCP clients on different networks communicating with DHCP server via relay agent.

DHCP Relay Packet Flow

- DHCP Discover and Request messages from clients are relayed to the DHCP server.
- Server responses (Offer, Ack) are relayed back to clients.

Activity: Troubleshooting DHCP

- Steps to troubleshoot DHCP issues:
    - Verify physical connectivity.
    - Check DHCP server status and scope availability.
    - Confirm relay agent configuration.
    - Use commands like
    - `ipconfig /all`
    - or
    - `dhclient`
    - to check IP address status.
    - Review logs on DHCP server.

Lesson 6.5: Domain Naming System (DNS)

What is DNS?

- Translates human-friendly domain names (e.g., comptia.org) into IP addresses.
- Essential for internet and network usability.

DNS Hierarchy

- Root Domain (top-level)
- Top-Level Domains (TLDs), e.g., .com, .org
- Domain and Sub-Domains

Name Resolution Using DNS

- Process of resolving domain names to IP addresses to enable communication.

Resource Records in DNS

Record Type	Purpose
SOA	Start of Authority; zone authority and admin info
A	Maps domain names to IPv4 addresses
AAAA	Maps domain names to IPv6 addresses
MX	Specifies mail exchange servers for a domain
SRV	Specifies service locations including ports
TXT	Stores text info for external sources (e.g., SPF, DKIM)
PTR	Pointer record for reverse DNS lookup (IP to domain)

DNS Server Configuration

- Can be configured for internal network use or external public DNS services.

Internal vs. External DNS

- Internal DNS resolves names within a private network.
- External DNS resolves public domain names on the internet.

DNS Security Threats and Solutions

Threat	Solution
DNS Spoofing/Poisoning	DNSSEC (DNS Security Extensions)
Eavesdropping/Interception	DNS over TLS (DoT)
Tampering	DNS over HTTPS (DoH)
DNS Tunneling/Advanced Attacks	Advanced DNS Firewalls

Activity: Two Truths and a Lie

- Engaging exercise to identify true and false statements about DNS security.

Lesson 6.6: DNS Troubleshooting

Common Client DNS Issues

- Incorrect DNS server settings.
- Cached old or incorrect DNS entries.
- Network connectivity problems.

Name Resolution Issues

- Failures in translating domain names to IP addresses.
- Can be caused by DNS server problems or misconfigurations.

Activity: Run nslookup and dig

- On Windows: Use
- `nslookup comptia.org`
- in Command Prompt.
- On Linux/Mac: Use
- `dig comptia.org`
- in Terminal.
- These commands help diagnose DNS resolution issues.

Module Summary: Implementing Network Services

- **TCP:** Reliable, connection-oriented transport protocol with error checking.
- **UDP:** Connectionless transport protocol; faster but no guarantees.
- **DHCP:** Automates IP addressing and network configuration.
- **DHCP Relay Agents:** Forward DHCP traffic between subnetworks.
- **DNS:** Translates domain names to IP addresses using a global hierarchical system.

Vocabulary List:

ACK (Acknowledgment)
A signal sent in TCP communication to confirm receipt of data.

APIPA (Automatic Private IP Addressing)
A feature that assigns an IP address in the 169.254.x.x range when DHCP is unavailable.

BootP (Bootstrap Protocol)
A legacy protocol for automatic IP address assignment, predecessor to DHCP.

DHCP (Dynamic Host Configuration Protocol)
A network protocol that automatically assigns IP addresses and other network configuration to devices.

DHCP Discover
The initial broadcast message sent by a client to locate available DHCP servers.

DHCP Offer
A message from a DHCP server offering an IP address to a client.

DHCP Request
A message from the client requesting the offered IP address.

DHCP Acknowledgment (Ack)
The server's confirmation that the IP address is assigned to the client.

DHCP Relay Agent
A device or service that forwards DHCP messages between clients and servers across different subnets.

DNS (Domain Name System)
A hierarchical system that translates domain names into IP addresses.

DNSSEC (DNS Security Extensions)
Security protocol that adds authentication to DNS responses to prevent spoofing.

DoH (DNS over HTTPS)
A protocol for performing DNS resolution via encrypted HTTPS connections.

DoT (DNS over TLS)
A protocol that encrypts DNS queries and responses using TLS for security.

IP Address
A unique numerical label assigned to each device on a network.

IPv6
The most recent version of the Internet Protocol, providing a larger address space than IPv4.

LDAP (Lightweight Directory Access Protocol)
A protocol for accessing and maintaining distributed directory information services.

MX Record (Mail Exchange Record)
DNS record specifying mail servers responsible for accepting email for a domain.

Port Number
A numerical identifier in networking used to specify particular processes or network services.

POP (Post Office Protocol)
An email protocol used to retrieve messages from a mail server.

RDP (Remote Desktop Protocol)
A protocol that allows users to connect to another computer over a network connection.

SLAAC (Stateless Address Autoconfiguration)
An IPv6 feature that allows devices to self-configure an IP address without a DHCP server.

SMTP (Simple Mail Transfer Protocol)
Protocol for sending email messages between servers.

SNMP (Simple Network Management Protocol)
Protocol used for network management and monitoring.

SOA Record (Start of Authority)
DNS record that designates the authoritative information about a DNS zone.

SRV Record (Service Record)
DNS record that specifies the location of services within a domain.

SSH (Secure Shell)
A protocol for secure remote login and other secure network services.

TCP (Transmission Control Protocol)
A connection-oriented, reliable transport layer protocol.

UDP (User Datagram Protocol)
A connectionless, unreliable transport layer protocol.

URL (Uniform Resource Locator)
The address used to access resources on the internet.

Web Server (HTTP/HTTPS)
A server that delivers web content using HyperText Transfer Protocol or its secure variant.

Acronym List:

- **ACK** — Acknowledgment
- **APIPA** — Automatic Private IP Addressing
- **BOOTP** — Bootstrap Protocol
- **DHCP** — Dynamic Host Configuration Protocol
- **DNS** — Domain Name System
- **DNSSEC** — Domain Name System Security Extensions
- **DoH** — DNS over HTTPS
- **DoT** — DNS over TLS
- **HTTP** — HyperText Transfer Protocol
- **HTTPS** — HyperText Transfer Protocol Secure
- **IMAP** — Internet Message Access Protocol
- **IP** — Internet Protocol
- **IPv4** — Internet Protocol version 4
- **IPv6** — Internet Protocol version 6
- **LDAP** — Lightweight Directory Access Protocol
- **MX** — Mail Exchange
- **NTP** — Network Time Protocol
- **POP** — Post Office Protocol
- **RDP** — Remote Desktop Protocol
- **RTCP** — Real-Time Control Protocol
- **RTP** — Real-Time Protocol
- **SIP** — Session Initiation Protocol
- **SMTP** — Simple Mail Transfer Protocol
- **SNMP** — Simple Network Management Protocol
- **SOA** — Start of Authority
- **SRV** — Service Record
- **SSH** — Secure Shell
- **SSL** — Secure Sockets Layer
- **TCP** — Transmission Control Protocol
- **TFTP** — Trivial File Transfer Protocol
- **TLS** — Transport Layer Security
- **UDP** — User Datagram Protocol
- **URL** — Uniform Resource Locator
- **SMTPs** — SMTP Secure (SMTP over SSL/TLS)
- **IMAPS** — IMAP Secure
- **POP3s** — POP3 Secure
- **SMB** — Server Message Block